



A Gentle Introduction to Reverse Mathematics

J.V. Siratt

Department of Mathematics
University of Notre Dame

October 27, 2021

Needed Background

- I assume a basic familiarity with logic, computability theory, and model theory, such as what is covered in our first year sequence.
- We will be making heavy use of Post's Theorem and $\Sigma/\Pi/\Delta$ notation for quantifier complexity.
- Where possible I will sketch out proofs using these basic ideas.
- Where the above is not possible, I will shamelessly handwave.

What is Reverse Math?

Math:

From these axioms we prove this theorem.

Backwards Math?:

From this theorem we prove we need these axioms?

Give a language $\mathcal{L}$ and a base theory T , we can define an ordering on $\mathcal{L}$ -sentences:

$$\psi \preceq \phi \text{ if } T \vdash \phi \rightarrow \psi$$

(or equivalently $(T + \phi) \vdash \psi$)

Second Order Arithmetic

Specifically, we are interested in the language of second order arithmetic L_2 .

We need a base theory to work in that:

- Is powerful enough to prove “useful” tools exist.
- Is weak enough that it is independent of “interesting” sentences.

If it's too weak then we will struggle to prove anything.

If it's too strong then everything will have equivalent strength.

We need it to be just right.

Second Order Arithmetic

Basic Axioms:

- $n + 1 \neq 0$
- $m + 1 = n + 1 \rightarrow m = n$
- $m + 0 = m$
- $m + (n + 1) = (m + n) + 1$
- $m \cdot 0 = 0$
- $m \cdot (n + 1) = (m \cdot n) + m$
- $\neg m < 0$
- $m < n + 1 \leftrightarrow (m < n \vee m = n)$

Essentially a modification of Robinson Arithmetic (Q).

Second Order Arithmetic

Axioms of Full Second Order Arithmetic (Z_2):
Basic + Full Comprehension + Set Induction

From Z_2 one can derive almost all of classical mathematics.

This is way too strong, so we will be interested in subsystems of second order arithmetic.

A common theme in reverse mathematics is to consider weaker set existence axioms:

- restricted comprehension
- existence of particular families of sets

Structures in Second Order Arithmetic

We treat second order arithmetic as a two-sorted first order system.

For clarity, we call the first sort the “first order” part, etc.

An ω -model has the standard numbers as its first order part.

We may refer to these simply by their second order part.

The intended model is of course $(\omega, \mathcal{P}(\omega))$.

An arbitrary model the first order part may be a superset of ω .

The second order part of an ω -model may be a subset of $\mathcal{P}(\omega)$.

Set Existence and Induction

$Z_2 = \text{Basic} + \text{Full Comprehension} + \text{Set Induction}$

The axioms of Z_2 establish that all analytical (second order definable) sets exist.

With set induction, this gives us induction over all formulas in L_2 .

An obvious restriction then would be restricting the comprehension axiom to arithmetical (first order definable) sets.

This gives us the **Arithmetical Comprehension Axiom** or **ACA**.

ACA with set induction gives us induction over all arithmetical formula:

An arithmetical formula with arithmetical set parameters is still arithmetical.

We use subscript 0 to denote restricted induction, so:

ACA₀ = Basic + Arithmetical Comprehension + Set Induction

Set Existence and Induction

In fact:

$$\text{ACA}_0 = \text{Basic} + \Sigma_1^0\text{-comprehension} + \text{Set Induction}$$

Proof sketch: Σ_1^0 -comprehension means we can construct the halting set K . All sets computable from K can then be constructed by comprehension since they are Δ_0^0 with respect to K . Σ_1^0 -comprehension then allows us to construct the jump of each of these, etc.



So if we want to further restrict comprehension, we have to go lower than Σ_1^0 .

Set Existence and Induction

We consider then the **Recursive Comprehension Axiom** or **RCA**, or Δ_1^0 -comprehension.

Again, along with set induction this would give us Δ_1^0 -induction.

But lets strengthen that slightly and give ourselves a bit of wiggle room:

$$\mathbf{RCA}_0 = \text{Basic} + \Delta_1^0\text{-comprehension} + \Sigma_1^0\text{-induction}$$

The Big Five

Traditionally, the main subsystems of Z_2 given in order of increasing strength are:

- RCA_0
- WKL_0
- ACA_0
- ATR_0
- $\Pi^1_1\text{-}CA_0$

You can possibly guess that $\Pi^1_1\text{-}CA_0$ denotes Π^1_1 -comprehension with restricted induction.

WKL_0 and ATR_0 arise somewhat less obviously, and involve strengthening the previous system with an assertion that a particular type of set exists.

The Big Five

Our focus will be on the first three:

- $\text{RCA}_0 = \text{Basic} + \Delta_1^0\text{-comprehension} + \Sigma_1^0\text{-induction}$
- $\text{WKL}_0 = \text{RCA}_0 + \text{WKL}$
- $\text{ACA}_0 = \text{Basic} + \text{Arithmetic Comprehension} + \text{Set Induction}$

Where WKL is Weak König's Lemma,
Every infinite binary tree has a path.

We will circle back to this later, but for now we will further investigate RCA_0 and ACA_0 .

First Order Parts

$PA = \text{Basic} + \text{Arithmetic Induction}$

$I\Sigma_1 = \text{Basic} + \Sigma_1\text{-induction}$

Here the induction axioms are strictly first order.

Their second order counterparts extend to allow set parameters.

In ACA_0 , arithmetic comprehension and set induction give us arithmetic induction (with parameters), so any model of ACA_0 satisfies PA with its first order part.

RCA_0 explicitly gives us Σ_1^0 -induction, so any model of RCA_0 satisfies $I\Sigma_1$ with its first order part.

We ask, do the converses hold?

That is, can we extend any model of $PA/I\Sigma_1$ to a model of ACA_0/RCA_0 ?

First Order Parts

The first order part of ACA_0 is precisely PA.

Proof sketch: Take a model of PA and construct a second order part consisting of the arithmetically definable sets. This satisfies comprehension.

Given any arithmetical formula with arithmetical set parameters, we can replace the sets with their formula and obtain an arithmetical formula without parameters.

So arithmetical comprehension doesn't give us “more” induction than what is allowed in PA – that is, we don't have to modify the first order part to model ACA_0 .



First Order Parts

The first order part of RCA_0 is precisely $\text{I}\Sigma_1$.

Proof sketch: Take a model of $\text{I}\Sigma_1$ and construct a second order part consisting of the Δ_1^0 definable sets to satisfy comprehension.

A Σ_1^0 formula with Δ_1^0 set parameters is still Σ_1^0 , so we have a model of RCA_0 .



Second Order Parts of ω -models

The second order parts of ω -models of RCA_0 and ACA_0 are respectively characterized by the following:

A **Turing Ideal** is a subset of ω which is closed under Turing reduction and the computable join.

A **Jump Ideal** is a Turing Ideal that is additionally closed under the Turing jump.

Second Order Parts of ω -models

The ω -models of RCA_0 are precisely the Turing ideals.

Proof sketch: $\Rightarrow$ Let M be an ω -model of RCA_0 .

If Z is computable from $X \in M$, then Z is Δ_1^0 definable with X as a parameter. Hence $Z \in M$.

For $X, Y \in M$, their join is computable in those parameters, so it is also Δ_1^0 definable, completing the closure.

$\Leftarrow$ Let N be (ω, I) where I is a Turing ideal.

Let ϕ be Δ_1^0 with parameters $X_0, \dots, X_n \in I$. The serial join $\overline{X}$ of $X_0, \dots, X_n$ is in I .

The set $Y = \{n \mid \phi(n)\}$ is then computable from $\overline{X}$, so $Y \in I$.

So N is closed under Δ_1^0 definability.



Second Order Parts of ω -models

The ω -models of ACA_0 are precisely the jump ideals.

Proof sketch: $\Rightarrow$ Let M be an ω -model of ACA_0 .

Since ACA_0 extends RCA_0 , M must be a Turing ideal. It suffices to show closure under the jump.

For any $X \in M$, X' is Σ_1^0 in X , so arithmetic comprehension gives us $X' \in M$.

$\Leftarrow$ Let N be (ω, I) with I a jump ideal.

Let ϕ be arithmetical with parameters $X_0, \dots, X_n \in I$. Since I is an ideal, it contains the join $\overline{X}$ of $X_0, \dots, X_n$.

The set $Y = \{n \mid \phi(n)\}$ is then a Σ_k^0 for some k and computable in the k th jump of $\overline{X}$. So $Y \in I$.

□

Minimal ω -models

In addition to characterizing the ω -models of RCA_0 and ACA_0 , these results also provide us with minimal ω -models.

$\text{REC} = \{X \mid X \text{ is computable}\}$

is the minimal Turing ideal, and so the minimal ω -model of RCA_0 .

$\text{ARITH} = \{X \mid X \text{ is arithmetic}\}$

is the minimal jump ideal, and so the minimal ω -model of ACA_0 .

This connection between RCA_0 and the computable sets, and between ACA_0 and the relatively computable sets, allows us the convenience of using tools from computability theory in establishing proof-theoretic results.

Proof-theoretic Strength

Previously: Z_2 proves most classical mathematics, so we need a weaker base system.

RCA_0 has been show to prove a number of things:

- a weak version of Gödel's completeness theorem
- the existence (but not uniqueness) of an algebraic closure of a countable field
- Baire category theorem (involves complete separable metric spaces and dense subsets)
- Banach/Steinhouse uniform boundedness theorem (involves linear functionals on separable Banach spaces)

There are many more good examples.

Proof-theoretic Strength

Over RCA_0 , it has been shown that ACA_0 is equivalent to:

- every Cauchy sequence is convergent in any complete separable metric space
- every countable integral domain/commutative ring has a maximal ideal
- every countable vector space over a countable field has a basis
- (Full) **König's Lemma** (every infinite finitely-branching tree has a path)
- Ramsey's Theorem, specifically: RT_k^n , for all k and all $n \geq 3$

RT_k^n is the statement:

For every k -coloring of size n subsets of ω , there exists an infinite monochromatic (homogeneous) subset of ω .

König's Lemma and Ramsey's Theorem will be important later.

Choosing a Base System

So RCA_0 is strong enough to prove some useful things, but the equivalences of ACA_0 show there are still interesting statements independent of RCA_0 .

So why don't we go even lower?

There has been investigation of weaker theories, such as:

$$\text{RCA}_0^* = \text{Basic} + \Delta_1^0 \text{ Comp} + \Delta_0^0 \text{ Ind} + \text{Exp Is Total}$$

We lose totality of exponentiation with the additional restriction to induction, so we need it as an axiom.

Things start to get messy.

Choosing a Base System

To illustrate this, in RCA_0 , TFAE:

- a subset has the same cardinality as $\mathbb{N}$
- a subset is unbounded.

In RCA_0^* , these are not equivalent, so the idea of infinity can be a bit slippery.

RCA_0 provides a good compromise by providing enough strength to reason somewhat traditionally, while still being weak enough to provide meaningful distinctions in proof-theoretic strength.

The Road Thus Far

	RCA_0	ACA_0
Comprehension	Δ_1^0	Arithmetical
Induction	Σ_1^0	Arithmetical
First Order	$I\Sigma_1^0$	PA
ω -models	Turing Ideals	Jump Ideals
Minimal	REC	ARITH

Even if we knew nothing else, we could reason:

Since ACA_0 extends RCA_0 but $REC \subsetneq ARITH$, RCA_0 is strictly weaker than ACA_0 .

We will write $RCA_0 < ACA_0$.

In general, we write $\Gamma_1 \leq \Gamma_2$ when $RCA_0 + \Gamma_2 \vdash \Gamma_1$, with $<$ and $=$ defined accordingly.

Leveraging Computability

Let Γ be some collection of L_2 sentences.

If $\text{RCA}_0 + \Gamma$ proves a non-computable set must exist, then it follows that $\text{RCA}_0 < \Gamma$.

If $\text{RCA}_0 + \Gamma$ can be satisfied while avoiding closure under the jump, then it follows that $\Gamma < \text{ACA}_0$.

So there is a very close relation between the computability-theoretic strength of many statements and their proof-theoretic strength.

Many results from computability theory therefore convert directly into results in reverse mathematics.

WKL₀

$$\text{WKL}_0 = \text{RCA}_0 + \text{WKL}$$

where WKL is the statement:

All infinite binary trees have a path.

$$\text{WKL}_0 \leq \text{ACA}_0.$$

Proof sketch: Let T be an infinite binary tree.

With arithmetic comprehension we can define T^* as every element of T which has infinitely many extensions in T :

$$\{\sigma \in T \mid \forall n > |\sigma|, \exists \tau \in T, |\tau| = n \wedge \sigma \prec \tau\}.$$

So T^* is an infinite binary tree with no dead ends.

We can construct a path essentially by starting at the base of T^* and choosing the leftmost branch every time a choice is offered (this process is primitive recursive).



WKL₀

Two quick notes:

- A benefit of using RCA_0 as our base system is that it allows us to make primitive recursive constructions such as building p from T^* in the previous proof.
- A similar proof shows one side of the equivalence of full König's Lemma and ACA_0 .

So we now have:

$$\text{RCA}_0 \leq \text{WKL}_0 \leq \text{ACA}_0$$

We can show that WKL_0 is strictly between the two using results from computability.

WKL₀

$$\text{WKL}_0 < \text{ACA}_0$$

Proof sketch: Recall:

- A set X is low if $X' \leq_T 0'$. A set X is low relative Y if $X' \leq_T Y'$.
- The Low Basis Theorem states that every computable infinite binary tree has a low path.

WKL₀ extends RCA₀, so it's ω -models are Turing ideals.

Given $X \in \text{REC}$, there then exists a low solution Y .

Relativizing low basis theorem, Y has a solution Z low relative Y .

Z itself is then low since $Z' \leq_T Y' \leq_T 0'$.

This implies that we can extend REC with low sets to obtain an ω -model of WKL₀.

A model with only low sets is not closed under the jump, so it can not satisfy ACA₀.

WKL₀

$$\text{RCA}_0 < \text{WKL}_0$$

Proof sketch: Recall:

- Sets A and B are computably separable if there exists computable C such that $A \subseteq C$ and $B \cap C = \emptyset$.
- There exist (disjoint) computably inseparable c.e. sets.
- For a c.e. set A , we have computable sets A_s consisting of the elements enumerated by step s .

WKL₀

Proof sketch: (Continued)

For any two disjoint c.e. sets A and B , we can define a computable infinite binary tree $T = \bigcup T_n$ where:

$$T_n = \{\sigma \in 2^n \mid \forall i \leq n, i \in A_n \rightarrow \sigma(i) = 1 \wedge i \in B_n \rightarrow \sigma(i) = 0\}.$$

By construction, any path through T is the characteristic function of a set separating A and B .

Taking A and B to be computably inseparable, we have a computable T with no computable path.

It follows then that REC does not satisfy WKL₀.



Characterizing Models of WKL_0

Characterizing the first order parts of models and the ω -models of WKL_0 is more difficult than doing so for RCA_0 and ACA_0 .

The first order part of WKL_0 is $I\Sigma_1$ – the same as RCA_0 .
This follows directly from the following conservation result:

- WKL_0 is Π_1^1 conservative over RCA_0 –
that is, for any ψ of form $\forall X_1, \dots, X_n, \phi$, where ϕ is arithmetical, $WKL_0 \vdash \psi$ if and only if $RCA_0 \vdash \psi$.

Of course one direction is trivial.

The other direction requires a forcing argument.

Characterizing Models of WKL_0

ω -models of WKL_0 are more interesting – they are called Scott sets and have a close relationship with PA degrees.

Recall: A set X has PA degree if it computes a (consistent) completion of PA.

This is a deep topic, so we limit ourselves:

- Since there is a computable tree for which each path is a complete and consistent extension of PA, WKL_0 implies the existence of a set of PA degree.
- (Scott Basis Theorem) Every computable infinite binary tree has a path with PA degree.
- (Scott) Let X be a set of PA degree. There exists an ω -model of WKL_0 consisting entirely of X -computable sets.

A Brief Intermission

ACA₀
|
WKL₀
|
RCA₀

Within the Big Five

We have already given examples of what is provable in RCA_0 and what is equivalent to ACA_0 .

Over RCA_0 , the following are equivalent to WKL_0 :

- Gödel's completeness theorem
- every countable field has a unique algebraic closure
- Brouwer's Fixed Point Theorem (continuous functions on convex subsets of $\mathbb{R}^n$ have a fixed point)
- Hahn/Banach for separable Banach spaces (bounded linear functionals on a subspace have an extension)

Many important results are equivalent to one of the Big Five.

Not everything plays nicely, though.

Ramsey's Theorem

Define $[X]^n$ to be the subsets of X with size n .

We will often refer to these as (unordered) n -tuples.

A function $f : [\mathbb{N}]^n \rightarrow k$ is called a coloring.

A set $I \subset \mathbb{N}$ for which a coloring f is constant is called monochromatic (or homogeneous).

Ramsey's Theorem (RT):

Let $n, k \in \mathbb{N}$. For all k -colorings of n -tuples, there exists an infinite monochromatic set.

RT_k^n is the formal statement of RT in L_2 for fixed n, k .

Ramsey's Theorem

For fixed k , RCA_0 proves:

- RT_k^1 (pigeonhole principle)
- $\text{RT}_k^{n+1} \rightarrow \text{RT}_k^n$
- $\text{RT}_k^n \rightarrow \text{RT}_{k+1}^n$
- ACA_0 is equivalent to RT_2^3 (in fact, RT_k^n for $n \geq 3$)

These results give us

$$\text{RCA}_0 \leq \text{RT}_2^2 \leq \text{ACA}_0$$

Similar to the case for WKL_0 , this can be strengthened.

$$\text{RCA}_0 < \text{RT}_2^2$$

Proof sketch: (Jockusch '72)

$$\text{Define } h_i(n, s) = \begin{cases} \Phi_{i,s}^{K_s}(n) & \text{if defined} \\ 0 & \text{o.w.} \end{cases}$$

For $X \leq_T 0'$, exists an e s.t. $X = \Phi_e^K$. So $X(n) = \lim_s h_e(n, s)$.

Construct computable coloring c . Step s :

- substep $e < s$,
 - if $h_e(n, s) = 1$ for at least $2e + 2$ numbers $n < s$,
take the least $x < y < s$ for which $h_e(x, s) = h_e(y, s) = 1$ and
define $c(x, s) = 0$ and $c(y, s) = 1$,
 - otherwise, do nothing.
- For substep $e = s$, for all $n < s$ such that $c(n, s)$ is not yet
defined, let $c(n, s) = 0$.

RT₂²

Proof sketch: (Continued)

For any infinite $X \leq_T 0'$, we have $X(n) = \lim_s h_e(n, s)$ for some e . That is, $h_e(n, s) = X_s(n)$.

Consider the first $2e + 2$ elements of X . There exists a $k' > e$ s.t. for each $j \geq k'$, $h_e(m, j) = 1$ for each m in this first $2e + 2$ elements.

Since X is infinite, there is $k > k'$ s.t. additionally we have $k \in X$.

By construction, at step k substep e we then took $x < y$ of these $2e + 2$ elements and defined the color of (x, k) and (y, k) to be different.

So X can not be a homogeneous set for c .

Since there is no computable infinite monochromatic set, REC can not model RT₂².

RT_2^2

The previous proof actually shows more than $RCA_0 < RT_2^2$.

The coloring constructed does not have any infinite monochromatic sets computable in the jump, so RT_2^2 can not have a low model.

It follows then that WKL_0 does not imply RT_2^2 .

RT_2^2

Theorem. (Seetapun '95)

Let c be a coloring of pairs and $X \not\leq_T c$. There is an infinite monochromatic set H s.t. the join of c and H does not compute X .

Let $X = 0'$ in the above. Then any coloring that does not compute the jump has a solution that does not compute the jump.

This implies that we can construct a model that avoids the jump. (This is called cone avoidance.)

Since there is an ω -model that is not a jump ideal, $RT_2^2 < ACA_0$.

RT_2^2

In addition to the previous, we can take the following route to show there is a model that is not a jump ideal.

Theorem. (Cholak, Jockusch, and Slaman '01)

Every computable k -coloring of pairs has a low_2 infinite monochromatic set.

Similar to how we showed WKL_0 has a low model, we can relativize this to show a low_2 model satisfies RT_2^2 since low_2 relative a low_2 set is still low_2 .

RT_2^2

One final result brings us to a nice stopping point.

Theorem. (Liu '12)

Let c be a coloring of pairs. Then if c is not of PA degree, there exists infinite monochromatic H s.t. the join of c and H is not of PA degree.

It follows then that RT_2^2 does not imply WKL_0 .

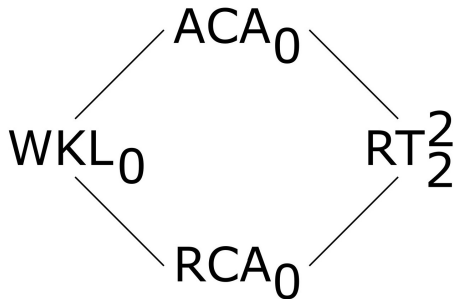
Note how this combines a form of cone avoidance with the fact that any ω -model of WKL_0 must include a set of PA degree.

Summary

So we have

- $\text{RCA}_0 < \text{WKL}_0 < \text{ACA}_0$
- $\text{RCA}_0 < \text{RT}_2^2 < \text{ACA}_0$
- RT_2^2 and WKL_0 are incomparable

Beautiful in its Simplicity?



Beautiful in its Simplicity?

